



Hardware Article



DIY Arduino calorimeter: Converting commercial thermally insulated coffee-cups into precision isobaric calorimeters

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ABSTRACT

We present the design, construction, and validation of a low-cost digital calorimeter intended for teaching thermochemistry in chemistry, engineering, and life sciences curricula. The device repurposes a commercial 380 mL thermal container with integrated magnetic stirring, transforming it into a closed and well-insulated constant-pressure system suitable for calorimetric experiments. Temperature monitoring is achieved with a waterproof Dallas DS18B20 digital sensor interfaced to an Arduino UNO board, allowing adjustable resolution and automated data acquisition. Open source software provides real time graphical visualization, CSV export, and optional Windows executables for straightforward deployment.

Compared to traditional educational calorimeters based on manual thermometry, this device offers improved reproducibility, elimination of reading errors, and enhanced student engagement through live data analysis. In contrast to professional-grade commercial calorimeters (US \$15,000–100,000) and basic educational kits (US \$50–500 without digital monitoring), our Arduino-based system (~US \$30 in components) provides an accessible middle ground that democratizes laboratory experimentation. Validation experiments, including neutralization and dissolution reactions, confirm reliable thermometric response, stable equilibrium detection, and enthalpy values consistent with literature.

This open-hardware calorimeter not only broadens access to thermochemistry education but also serves as a platform for interdisciplinary training in electronics, programming, and laboratory automation, fostering early exposure to modern research practices.

1. Specifications table

Hardware name	Digital calorimeter
Subject area	- Chemistry and biochemistry - Educational tools and open source alternatives to existing infrastructure - Measuring physical properties and in-lab sensors
Hardware type	

(continued on next page)

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(continued)

Hardware name	Digital calorimeter
Closest commercial analog	No low-cost commercial analog available for educational settings
Open source license	Creative Commons AttributionShareAlike 4.0 International License
Cost of hardware	USD 30
Source file repository	https://doi.org/10.17632/yvr5mvwbyv

2. Hardware in context

Many chemical reactions exchange measurable amounts of heat with their environment, and the study of this exchange is fundamental to understanding chemical processes, giving rise to the field of thermochemistry. This field is essential in the education of chemistry, engineering, and life sciences students [1–3]. Teaching thermochemistry requires a system that allows the reaction under study to be carried out under controlled conditions and thermally isolated from the external environment. Traditionally, this is accomplished in thermal containers in which the chemical reaction (system) takes place in an aqueous environment (surroundings) whose temperature is monitored using an analog thermometer. Traditionally, this requires manual temperature recording during continuous agitation, a process that can be tedious and that is prone to measurement errors due to irregular readings and inconsistent stirring. Additionally, traditional manual thermometry suffers from methodological limitations such as limited temporal resolution, susceptibility to reading and transcription errors, and lack of automated data logging [1]. While the historical use of mercury thermometers poses well-known safety concerns [4,5], even non-toxic alternatives such as alcohol thermometers do not overcome these fundamental limitations. Accidents from spilling reactants are also common when agitation is done manually. High-precision digital equipment exists that avoids the use of mercury thermometers, but their cost is prohibitive for educational settings, where multiple pieces of equipment are also needed to allow greater practical student participation. In response, multiple attempts have been made to build calorimeters based on thermal cups (coffee-cup type) and monitored by electronic sensors [6–8]. Some of these lack agitation, depend on relatively expensive materials or software, or employ an external aquarium pump for stirring [9]; the latter prevents the system from remaining closed and thus requires explicit consideration of mass and energy exchange with the environment in the calorimetric analysis.

From an experimental standpoint, the quality of thermometric measurements also depends on the temporal response of the measuring system, which is governed by both the temperature sensor and the stirring conditions. Adequate stirring promotes rapid homogenization of the solution and reduces local temperature gradients, while appropriate thermal coupling between the sensor and the liquid phase enables timely detection of temperature changes. Different transient behaviors can therefore be observed depending on how these factors are balanced in a given setup.

In the present work, a closed and well-insulated reaction vessel with integrated magnetic stirring was selected to address these considerations in a pragmatic manner. This design minimizes uncontrolled exchanges with the environment over the time scale of the experiment and provides a stable and reproducible thermometric response, while maintaining simplicity, low cost, and suitability for teaching and exploratory laboratory settings.

This study presents the construction and validation of a digital calorimeter obtained by repurposing a commercial thermally insulated container with integrated magnetic stirring and incorporating a digital temperature monitoring and data logging system based on an Arduino UNO programmable board. The design addresses four objectives: (i) reducing costs compared to commercial equipment, (ii) improving precision and reproducibility in educational experiments, (iii) promoting active learning through hands-on student engagement with assembly, calibration, and operation of the instrument, and (iv) enhancing accessibility for teaching laboratories. Accessibility is achieved through the use of low-cost, widely available components and open source hardware and software, making the device suitable for laboratories with limited resources.

Arduino-based calorimeters have recently been proposed as low-cost alternatives for teaching thermochemistry. For example, Prabowo et al. [2] developed the ChemDuino calorimeter to determine the enthalpy change of acid–base neutralization reactions, demonstrating that reliable values can be obtained using a simple DIY setup. However, their design relies on disposable paper cups and a tumbler, with manual mixing and exposed electronic components, limiting its robustness and reproducibility for repeated use in educational laboratories. In contrast, our device repurposes a commercial thermal container with integrated magnetic stirring, ensuring a closed system with better thermal insulation and homogeneous mixing. Furthermore, the electronics are housed in a 3D-printed enclosure, making the device more durable, while the modular architecture of our system allows the integration of additional sensors (pH, conductivity, pressure), expanding its applicability beyond simple neutralization experiments.

Several previous studies have implemented digital temperature sensors and programmable boards [2,6,7], demonstrating that good precision and accuracy can be achieved with these components. However, these designs require complete device construction and, in some cases, include elements such as screens that may limit their applicability. In our work, we chose to utilize a commercial thermal container that included magnetic stirring and developed open source software for data acquisition and equipment control, connecting it via USB cable to a computer with Linux or Windows operating system for data visualization and subsequent post-processing.

High-precision digital calorimetric equipment that avoids the risks associated with mercury thermometers does exist, but costs are prohibitive for teaching laboratories. Professional-grade systems such as the IKA C 1 Package (ca. US \$18,000–20,000), the IKA C 6000 series (ca. US \$55,000–65,000), and calorimeters from Parr Instrument Company or TA Instruments, typically ranging from US \$15,000 to over US \$100,000 depending on model and capabilities, offer excellent precision and automation. However, these instruments remain inaccessible to most educational institutions that require multiple units for student laboratory sessions.

At the other extreme, basic educational calorimeter kits are available in the range of US \$50–500. These include simple “coffee-cup” calorimeters or student laboratory sets widely marketed for introductory chemistry education. While affordable, such kits lack digital monitoring, automated data acquisition, and precise temperature control, limiting their usefulness for modern experimental training. More recent educational developments have also explored reusable alternatives to traditional coffee-cup calorimeters, such as low-cost 3D-printed containers with improved system closure and durability, while maintaining the accessibility and pedagogical focus required for teaching laboratories [8].

This creates a significant gap between rudimentary educational tools and professional instrumentation—a gap directly addressed by our US \$30 Arduino-based digital calorimeter, which combines affordability with digital data acquisition and reproducibility suitable for classroom use.

A comparison between the present calorimeter and representative commercial isobaric calorimeters is shown in Table 1.

Commercial solution calorimeters, such as the Parr 6755, employ precision thermistors and achieve high temperature resolution and accuracy after calibration. Data acquisition in these instruments relies on digitally logged measurements with fixed logging intervals defined by the instrument software. Educational kits, such as the Vernier Go Direct Temperature Probe, use a thermistor-based digital probe and a foam-cup calorimeter. In this context, a complete experimental setup requires additional components, including a thermally insulated container and a stirring system, which are already integrated in the proposed device.

For the developed device, the implemented sampling rate of approximately 1 Hz is sufficient to resolve temperature transients associated with typical dissolution and neutralization reactions.

The main limitations of the present calorimeter are related to its lower temperature resolution and simpler thermal insulation when compared to commercial instruments. Consequently, the device is not intended to replace research-grade calorimeters designed for high-precision or microcalorimetric measurements. Nevertheless, the validation results demonstrate that, after calibration, the system provides reproducible and quantitatively meaningful results for isobaric processes involving moderate heat exchange, making it suitable for educational and low-cost experimental laboratory settings.

Low-cost calorimeters not only democratize access to thermal experimentation, but also serve as vehicles for teaching electronics, programming, and experimental design in an interdisciplinary context. Our open-architecture device allows combining the teaching of both the physical chemistry principles involved in thermochemical determinations and the analytical principles it requires, as well as teaching laboratory automation in the sciences. We believe this training is necessary and increasingly needs to be done earlier due to the growing automation of tasks in research laboratories [10,11]. The combination of these qualities fosters students' creative capacity to develop new ways of executing their experiments. This is evidenced by the growing number of studies aimed at incorporating the use of programmable boards, such as Arduino, in natural science education [12–17].

Table 1

Comparison between the present DIY isobaric calorimeter and representative commercial calorimeters.

Feature	Present calorimeter (DIY; DS18B20-based)	Vernier Go direct temperature probe	Parr 6755 solution calorimeter	TAM IV precision solution calorimeter (SolCal)
Calorimetric mode	Isobaric (constant pressure)	Not applicable (temperature sensor only; insulated container and stirring system not included)	Isobaric (solution calorimeter)	Isobaric (precision solution calorimeter)
Instrument type	DIY, open-hardware calorimeter developed by the authors	Commercial educational temperature sensor	Commercial laboratory solution calorimeter	Research-grade precision solution calorimeter
Temperature sensor	Digital thermometer DS18B20	Digital thermistor	Precision thermistor (Parr 6772)	High-stability thermistors
Temperature resolution	0.0625 °C (12-bit configuration)	0.01 °C	0.0001 °C	< 0.00001 °C (baseline noise < $\pm 10^{-5}$ °C)
Sampling/read frequency	1 Hz (software-configurable)	Not specified by manufacturer	≈ 0.1 Hz	Not specified by manufacturer; continuous data acquisition
Thermal insulation	Modified commercial insulated cup	Not included	Factory-designed insulated vessel	High-performance thermostated enclosure
Typical applications	Dissolution, neutralization, and mixing reactions with moderate heat exchange	Educational calorimetry experiments	Solution thermochemistry and routine laboratory measurements	High-precision dissolution and reaction calorimetry
Software ecosystem	Open source (Arduino + Python)	Proprietary	Proprietary	Proprietary
Approximate cost (US \$)	30	110 (sensor only)	>18,000	>15,000

Note: Technical specifications for commercial calorimeters were obtained from manufacturer manuals and technical datasheets (Parr Instrument Company; TA Instruments). Specifications for the Vernier Go Direct Temperature Probe were taken from product specifications and support documentation available on the Vernier website. Specifications for the DIY calorimeter are derived from the DS18B20 datasheet and from the experimental configuration described in this work. Approximate costs based on publicly available supplier information, online catalog listings, and typical manufacturer quotations at the time of manuscript preparation.

3. Hardware description

The developed calorimeter combines low-cost commercial components with digital data acquisition technology for educational calorimetric applications. The design is based on repurposing a commercial 380 mL thermal container as an adiabatic container, taking advantage of its inherent thermal insulation properties and its commercial availability at low cost. This approach fundamentally differs from traditional teaching calorimeters that require analog thermometers and manual data recording, as well as from professional systems that are prohibitively expensive for educational institutions.

The container used for equipment construction is a commercial thermal container with lid and magnetic stirring. This allows it to remain closed while the chemical reaction occurs inside.

The main hardware customization lies in the integration of a Dallas DS18B20 digital temperature sensor with an Arduino UNO board, creating an automated monitoring system that eliminates human reading errors and allows continuous data recording. The sensor is used in its waterproof stainless-steel encapsulated version, which places the sensing element in direct thermal contact with the liquid sample while providing mechanical and electrical protection. This configuration allows the sensor to be submerged in the reaction medium and positioned inside the calorimetric container, away from the container walls, to measure the temperature of the bulk solution. The electrical connections between the DS18B20 sensor and the Arduino UNO are deliberately simple and are illustrated schematically and photographically in Fig. 1. Specifically, the sensor is connected using a standard three-wire configuration: VCC to the Arduino 5 V pin, GND to ground, and the data line to a digital input pin (D2 in our implementation), with a 4.7 k Ω pull-up resistor placed between VCC and the data line.

A notable feature of the DS18B20 is its flexible configuration capability in terms of thermal resolution and response speed. The sensor allows selection between four different resolution levels: 9, 10, 11, and 12 bits, which correspond to precisions of ± 0.5 $^{\circ}\text{C}$, ± 0.25 $^{\circ}\text{C}$, ± 0.125 $^{\circ}\text{C}$, and ± 0.0625 $^{\circ}\text{C}$ respectively. This flexibility involves a trade-off between precision and response speed, as conversion times vary from 93.75 ms for 9-bit resolution to 750 ms for 12-bit resolution. This characteristic allows the sensor to be adapted according to specific experimental needs: high-speed configurations for rapid monitoring with lower precision, or high-precision configurations for critical measurements where accuracy is a priority.

For educational calorimeter applications, the DS18B20 presents significant advantages such as easy connection requiring only one digital pin from the Arduino, reading stability, configuration flexibility, low cost, and elimination of analog noise thanks to its digital interface. In our calorimeter, we use the default 12-bit configuration (± 0.0625 $^{\circ}\text{C}$, 750 ms) to obtain maximum available precision, although the conversion time is acceptable for typical thermal dynamics in educational calorimetry experiments.

The developed software that accompanies the device provides functionalities such as automatic data logging in CSV format, real time graphical visualization, and export capability for subsequent analysis.

The open source character of the design allows future adaptations according to different available containers, facilitating scalability and customization according to specific institutional resources. The modular architecture of the electronic system enables the

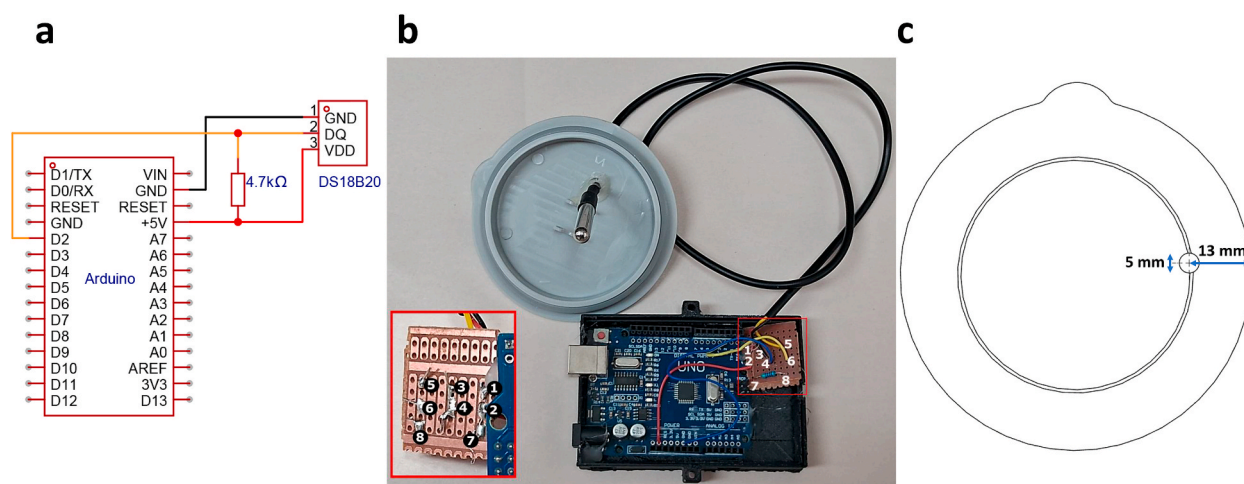


Fig. 1. Electrical connections and sensor installation details. (a) Electronic schematic of the Dallas DS18B20 temperature sensor connected to the Arduino UNO board, including power supply (5 V), ground (GND), data line (DQ), and the 4.7 k Ω pull-up resistor configuration. (b) Photograph of the assembled prototype circuit built on a perforated board and connected to the Arduino UNO. The main image shows the front view of the circuit with numbered connections, while the red inset shows the same board from the reverse side, highlighting the corresponding soldered connections. Numbers indicate: (1) DS18B20 VDD (5 V), (2) Arduino 5 V supply, (3) DS18B20 GND, (4) Arduino GND, (5) DS18B20 data line (DQ), (6) Arduino digital input pin (D2), (7, 8) connections to the 4.7 k Ω pull-up resistor. (c) Schematic view of the thermal container lid indicating the position and diameter of the perforation used for sensor insertion, located approximately 13 mm from the outer edge of the lid to avoid physical interference with the magnetic stirrer. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

incorporation of additional sensors (pH, conductivity, pressure) for more complex experiments, establishing an expandable platform for teaching-oriented scientific instrumentation. The integration of automatic magnetic stirring eliminates variability associated with manual stirring, improving experimental reproducibility compared to traditional methods.

Research Applications and Benefits:

- Automated thermochemical data acquisition: Enables continuous temperature monitoring during dissolution, neutralization, and crystallization reactions with sub-degree precision, eliminating manual reading errors and providing exportable datasets for advanced statistical analysis.
- Cost-effective laboratory modernization: Provides digital instrumentation capabilities to budget-constrained educational institutions, offering professional-grade data logging and visualization at a fraction of commercial alternatives, making modern analytical techniques accessible to underserved communities.
- Customizable platform for multiparameter sensing: The modular Arduino-based architecture can be readily expanded with additional sensors (pH, conductivity, pressure) for comprehensive reaction monitoring, enabling interdisciplinary research combining thermodynamics, kinetics, and analytical chemistry.
- Open source educational tool development: Serves as a foundation for developing institution-specific instrumentation, allowing researchers to modify hardware and software according to local curricula needs, experimental requirements, and available materials, promoting scientific instrument literacy.
- Real time experimental feedback: Provides immediate graphical visualization of thermal processes, enabling dynamic experimental adjustments and enhanced student understanding of thermodynamic principles through interactive data exploration and hypothesis testing during live experiments.

3.1. Software

The calorimeter software consists of two components: first, the Arduino UNO firmware (firmware.ino) that reports temperature readings from the Dallas DS18B20 sensor via the serial port; second, the PC-side scripts ([gui.py](#) and [calorimeter_support.py](#)), both written in Python and expected to reside in the same directory on the connected computer. For Linux and macOS systems, execution of the Python scripts requires a standard Python 3 environment and installation of the required third-party libraries. The required Python dependencies and step-by-step installation instructions are explicitly documented in the repository (see README and requirements file). This approach preserves cross-platform compatibility while keeping the manuscript concise. Optionally, these scripts can be executed in a Windows environment using the [calorimeter.exe](#) file, which is available in the repository along with the necessary dependencies for proper execution.

Prior to uploading the Arduino sketch, two external libraries must be installed: OneWire and DallasTemperature. These libraries can be installed directly from the Arduino IDE by navigating to Sketch → Include Library → Manage Libraries. First, search for and install the OneWire library, followed by the DallasTemperature library, using the Install button in the Library Manager.

The [calorimeter_support.py](#) file captures the readings received from the Arduino board through the serial port; the port to which the board is connected is automatically detected. The [gui.py](#) file generates a graphical user interface that allows operation of the calorimeter, including starting and stopping data acquisition, saving data in CSV format, navigating the graph, and clearing it ([Fig. 2](#)). Temperature readings are displayed in real time. The generated CSV files can subsequently be imported into spreadsheet software for data processing. Inline comments have been added to the Arduino firmware and Python scripts to clarify their structure and main functional blocks.



Fig. 2. Graphical user interface of the calorimeter: (1) Start/stop data acquisition, (2) Clear graph, (3) Save time and temperature data (CSV), (4) Reset zoom, (5) Current sensor temperature, (6) Detected COM port, (7) Exit interface.

The Dallas DS18B20 sensor offers adjustable configurations in both reading speed and measurement resolution. According to the manufacturer, this device can be configured with four different resolution levels: 9, 10, 11, and 12 bits, which correspond to precisions of ± 0.5 °C, ± 0.25 °C, ± 0.125 °C, and ± 0.0625 °C, respectively. The following conversion times correspond to the selected resolution:

- 9 bits: 93.75 ms.
- 10 bits: 187.5 ms.
- 11 bits: 375 ms.
- 12 bits: 750 ms.

In our specific case, the sensor's default configuration was used, which corresponds to a 12-bit resolution with a precision of ± 0.0625 °C and a conversion time of 750 ms. This configuration provides the maximum available precision of the sensor, although it requires longer processing time to complete each measurement; this time is acceptable for the calorimeter's intended operation. To select a different resolution, users can modify the Arduino firmware (firmware.ino) as indicated in the source code comments.

4. Design files summary

All design files associated with the calorimeter are released under a Creative Commons Attribution-ShareAlike 4.0 International (CC BY-SA 4.0) license. They include 3D-printable STL files for the Arduino case and adapter, Arduino firmware, and Python-based graphical user interface scripts. Executable versions for Windows are also provided for ease of deployment in teaching laboratories. All files have been deposited in the associated open-access repository (<https://doi.org/10.17632/yvr5mvwbyyv>) ensuring version control and long-term accessibility. These files enable straightforward replication of the device and provide a flexible basis for further modifications and extensions of the hardware.

Design file name	File type	Open source license	Location of the file
<i>arduino_case_bottom.stl</i>	STL file	Creative Commons AttributionShareAlike 4.0 International License	https://doi.org/10.17632/yvr5mvwbyyv
<i>arduino_case_top.stl</i>	STL file	Creative Commons AttributionShareAlike 4.0 International License	https://doi.org/10.17632/yvr5mvwbyyv
<i>arduino_adapter.stl</i>	STL file	Creative Commons AttributionShareAlike 4.0 International License	https://doi.org/10.17632/yvr5mvwbyyv
<i>arduino_adapter.step</i>	STEP file	Creative Commons AttributionShareAlike 4.0 International License	https://doi.org/10.17632/yvr5mvwbyyv
<i>firmware.ino</i>	Arduino firmware	Creative Commons AttributionShareAlike 4.0 International License	https://doi.org/10.17632/yvr5mvwbyyv
<i>calorimeter_support.py</i>	Python script	Creative Commons AttributionShareAlike 4.0 International License	https://doi.org/10.17632/yvr5mvwbyyv
<i>gui.py</i>	Python script	Creative Commons AttributionShareAlike 4.0 International License	https://doi.org/10.17632/yvr5mvwbyyv
<i>executable_packed.rar</i>	Compressed executable file for Windows system	Creative Commons AttributionShareAlike 4.0 International License	https://doi.org/10.17632/yvr5mvwbyyv
<i>README.md</i>	Markdown documentation file	Creative Commons AttributionShareAlike 4.0 International License	https://doi.org/10.17632/yvr5mvwbyyv
<i>requirements.txt</i>	Plain text dependency specification file	Creative Commons AttributionShareAlike 4.0 International License	https://doi.org/10.17632/yvr5mvwbyyv

arduino_case_bottom.stl and *arduino_case_top.stl*: 3D-printable parts forming the protective case for the Arduino UNO board.

arduino_adapter.stl: Adapter piece to connect the Arduino enclosure to the thermal mug.

arduino_adapter.step: Editable STEP (CAD) file provided to facilitate modification or re-modeling of the adapter geometry for alternative thermal containers.

firmware.ino: Arduino sketch that communicates with the DS18B20 sensor and sends temperature data via serial port.

calorimeter_support.py: Python script that reads serial data from the Arduino and formats it for visualization.

gui.py: Python graphical interface that displays temperature in real time and allows saving data as CSV.

executable_packed.rar: Pre-compiled Windows executable with all required dependencies for users without Python installed. After uncompressing this file, run *calorimeter.exe*. Note: Some antivirus software may incorrectly flag executables created with *cx_Freeze* or similar tools as suspicious due to their packaging method.

This is a false positive. If this occurs, please add the executable as an exception in your antivirus program to allow it to run.

README.md: Repository documentation describing software architecture, installation, hardware–software interaction, firmware upload, GUI execution, data outputs, and troubleshooting. Step-by-step guidance ensures reproducible operation on Linux, macOS, and Windows.

requirements.txt: Text file listing Python third-party libraries required for the calorimeter GUI and data acquisition software. It enables simple dependency installation via standard package managers, ensuring reproducible operation across systems.

5. Bill of materials summary

Designator	Component	Number	Cost per unit (USD)	Total cost (USD)	Source of materials
Item 1	Thermal mug with magnetic stirring	1	17	17	https://www.amazon.com ASIN: B0DSL76SXV
Item 2	Experimental PCB board	1	0.7	0.7	https://www.amazon.com ASIN: B00FXHXT80
Item 3	Electrical connection cables	3 colors, 10 cm each	0.3	0.9	Local hardware
Item 4	Generic Arduino UNO programmable board	1	7	7	https://www.amazon.com ASIN: B0BV9XFQ5T
Item 5	Electrical resistor (4.7 k Ω)	1	0.06	0.06	Local hardware
Item 6	Dallas DS18B20 temperature sensor (waterproof stainless-steel version)	1	2	2	https://www.amazon.com ASIN: B09NVFJYPS
Item 7	PLA filament	50 g	12/Kg	0.6	https://www.amazon.com B07XG3RM58
Item 8	USB Type A to Type B data cable	1	1.3	1.3	Local hardware
Item 9	Screws M3x12 mm	2	0.05	0.1	Local hardware
Item 10	Nuts M3	2	0.10	0.20	Local hardware

Direct hyperlinks are provided to the exact products used in this work. Product identifiers (Amazon ASINs, unique product identification codes) are included to unambiguously identify each component. Hyperlinks are used instead of full URLs to improve table readability; all links point to the exact products used.

The reported cost corresponds to the component prices at the time of device construction.

PLA filament cost is reported per kilogram; only a small fraction is consumed per device.

Supplies for soldering, hot glue, tools, and labor are not included in this budget quotation.

6. Build instructions

6.1. Selection of the thermal container

A commercial thermally-insulated mug with a capacity of 380 mL was selected as the base container (item 1), leveraging its thermal insulation properties and low-cost commercial availability. The specific thermally insulated mug used in this work corresponds to the model available at the time of device construction. As with many low-cost consumer products, commercial availability may vary over time. Importantly, the calorimeter design is not tied to a unique container model. The adapter components are provided as open-source

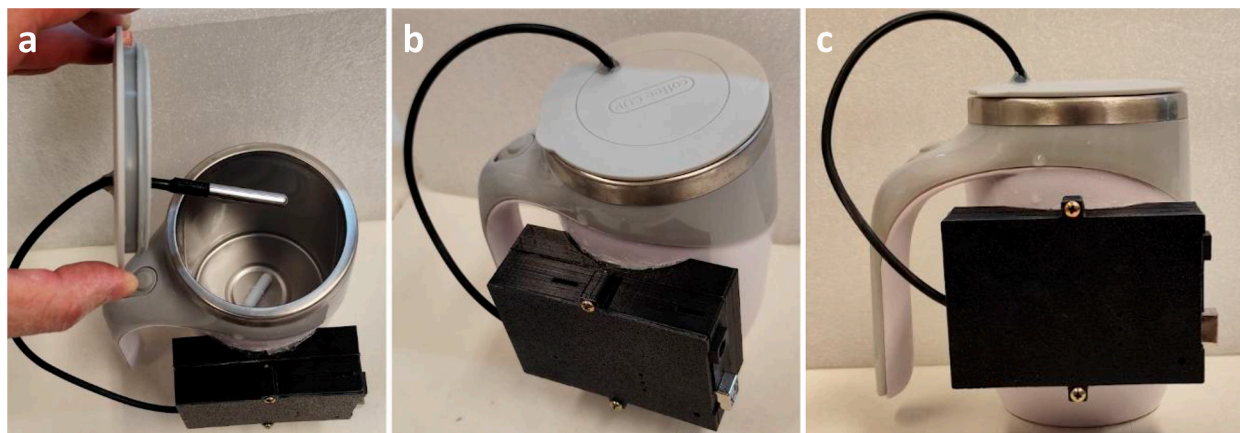


Fig. 3. (a) General internal and external appearance (b, c) of the calorimeter, using an Arduino UNO programmable board.

design files, including an editable STEP file. This allows users to readily adapt the geometry to alternative thermally insulated mugs of similar size and shape, thus ensuring long-term reproducibility of the device. The chosen model includes a built-in magnetic stirring system, which enables agitation while keeping the container physically closed and is rechargeable via a USB port. Equivalent commercially available self-stirring thermal mugs with similar capacity and insulation properties may be used.

6.2. Preparation of the container lid

The lid of the thermal container was perforated to allow insertion of the temperature sensor. The perforation was made at a position approximately 13 mm from the outer edge of the lid. This position was selected to avoid physical interference with the magnetic stirrer while ensuring that the sensing tip of the sensor remains fully immersed in the liquid during operation.

6.3. Installation and sealing of the temperature sensor

A waterproof Dallas DS18B20 temperature sensor (item 6) was inserted through the perforation in the lid. After positioning the sensor, the opening was sealed using hot-melt adhesive to mechanically secure the sensor and minimize vapor and heat exchange with the environment. The final position of the sensor and the perforation location are depicted in Fig. 1.

6.4. Electrical connections of the sensor

The DS18B20 sensor was connected to an Arduino UNO board (item 4) using a standard three-wire configuration. The sensor's VCC terminal was connected to the Arduino 5 V pin, the GND terminal to ground, and the data line to a digital input pin (D2 in our implementation). A 4.7 k Ω pull-up resistor (item 5) was placed between the VCC and data lines. These connections were implemented on an experimental PCB using standard electrical cables (items 2 and 3), as shown schematically and photographically in Figs. 1 and 3.

6.5. Assembly of the electronic enclosure

The Arduino UNO board and the experimental PCB were placed inside a 3D-printed enclosure manufactured from PLA filament (items 7, 9, and 10). The enclosure consists of a top and bottom case and an adapter piece designed to attach the enclosure to the thermal container. The STL files required to print these parts are provided in the associated repository.

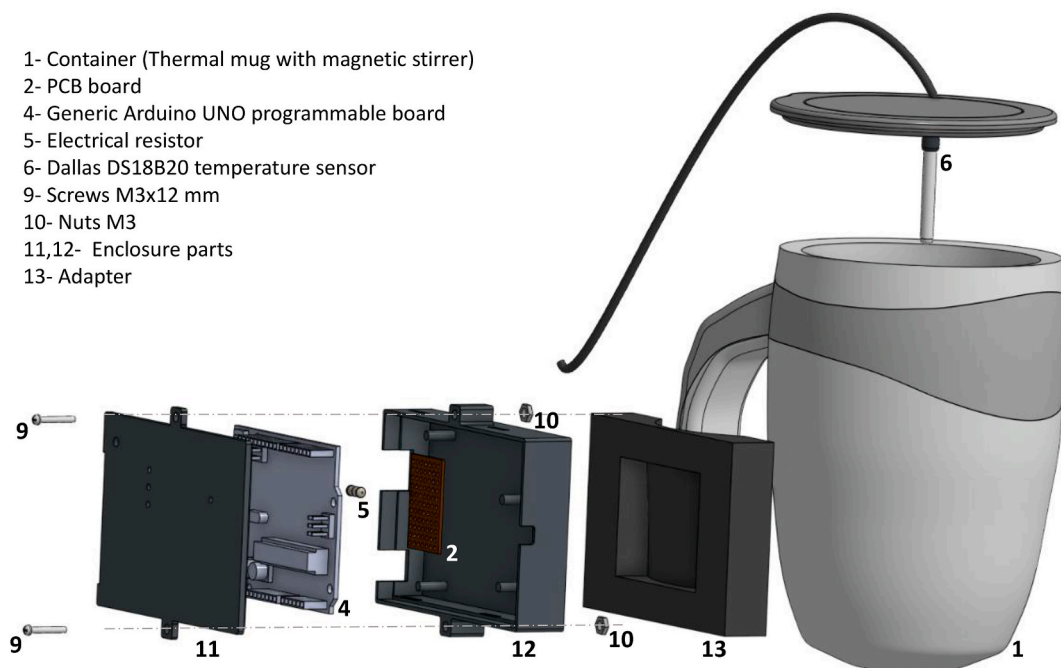


Fig. 4. Exploded assembly view of the 3D-printed enclosure and its mechanical integration with the thermal container, showing the assembly of the Arduino enclosure components and their attachment to the external wall of the thermal cup, indicating the relative position of the adapter (13), enclosure parts (11, 12), and container (1). The numbering of components corresponds to those summarized in the Bill of Materials table. For clarity of visualization, electrical connections are not shown in this figure; these are detailed separately in Fig. 1.

6.6. Attachment of the enclosure to the thermal container

The assembled enclosure was permanently attached to the external wall of the thermal container using hot-melt adhesive. Other equivalent fixing methods, such as epoxy adhesive or industrial double-sided tape, may also be used. An exploded assembly view illustrating the enclosure components and their attachment to the thermal container is provided in Fig. 4.

6.7. Connection to the computer

To connect the calorimeter to a personal computer, a USB Type B cable (item 8) was used. This type of connector provides a more stable mechanical connection when handled by students during laboratory sessions.

7. Operation instructions

1. For the first use of the calorimeter, connect the calorimeter to your personal computer's USB port using a USB Type B cable.
2. Download the software files stored in this article's repository and decompress them on the computer that will be used to operate the calorimeter.
3. Install the Arduino IDE software on your personal computer to upload the calorimeter firmware to the device. Open the firmware.ino file and upload it to the calorimeter's controller board using the "Upload" button in the Arduino IDE interface. Once this step is completed and no errors appear during the transfer, the firmware upload is finished and does not need to be repeated. Keep the calorimeter connected to the computer during normal operation.
4. Run the calorimeter operation program. This can be done from the command line using the Python interpreter installed on your system by executing the `gui.py` file. Note that this file must be located in the same directory as `calorimeter_support.py`. For Windows users, the program can also be run from the graphical environment by executing the `calorimeter.exe` file. This executable, together with the required dependencies, is included in the compressed archive `executable_packed.rar` deposited in the repository associated with this article.
5. A graphical user interface will open (Fig. 2). The software will automatically detect the COM port to which the calorimeter has been connected and display it in the lower left corner of the interface.
6. Pour the liquid in which the chemical reaction under study will take place, ensuring that the liquid level is above the temperature sensor so that it remains fully submerged.
7. Place the magnetic stir bar and activate the magnetic stirrer. If necessary, recharge the stirrer's battery by connecting a micro-USB cable to a USB port with 5 V output, which can be a computer's USB port or a mobile phone charger.
8. Place the lid on the calorimeter and start monitoring its internal temperature by pressing the "Start/Stop" button on the interface. Temperature values will begin to appear as they are being read.
9. Once the temperature has stabilized, briefly open the lid and add the second reactant to start the chemical reaction you wish to study.
10. Continue monitoring the temperature until it stabilizes again. The new stabilization value may be higher or lower than the initial temperature, depending on whether your chemical reaction is exo- or endothermic.
11. Stop data acquisition by pressing the "Start/Stop" button.
12. Save your data in CSV format by pressing the "Save data" button. This file can later be imported and analyzed with other software, e.g., using spreadsheet software.
13. The graphical interface allows you to use the mouse to select a region of the graph to zoom in on as well as navigate within it. To reset the zoom level, you can press the "Reset zoom" button.
14. You can clear the records displayed in the graph using the "Clear graph" button.
15. Exit the program by pressing the "Exit" button. Empty the calorimeter, then rinse its interior and the sensor housing.

8. Validation and characterization

A thermochemical process is a physical or chemical transformation accompanied by heat exchange with the environment, which can be followed by thermometry. A common example is the dissolution processes in which heat (Q) is absorbed (endothermic)/released (exothermic) from/to the environment. This exchange depends on the amount of substance, expressed as mass, m , or mole, n , and the heat capacity of the whole system (C_P), as defined in Eq. (1).

$$C_P = C_{CAL} + C_w \quad (1)$$

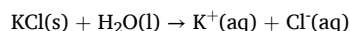
where C_w is the heat capacity of the water employed during the experiment, and C_{CAL} include all the pieces forming part of the calorimeter and that may also exchange heat, that is, the inner walls of the calorimeter, the temperature sensor, and the magnetic stir bar. These characteristics are specific to each calorimeter, and C_{CAL} should be determined for each device in a blank experiment under the same experimental conditions, including the time needed for the experiment to conclude in order to include any possible heat

exchange due to the solution agitation.

At constant pressure (e.g., atmospheric), the heat exchanged (Q_p) is equivalent to the enthalpy change: $\Delta_{sol}H^o = \int_{T_i}^{T_f} C_p dT$, in which the superscript “o” indicates the standard pressure of 1 Bar. If C_p is independent of temperature within the working temperature range, determining $\Delta_{sol}H^o$ involves only measuring ΔT of the process if C_{CAL} is known. The final equation to calculate the molar enthalpy of dissolution of a given substance X is:

$$\Delta_{sol}H_m^o(X) = \frac{-MW_X}{m_X} (C_{CAL} + m_w c_w) \Delta T \quad (2)$$

where m_w the experimental amount of water employed (in g), c_w is the specific heat capacity of water (4.184 J g^{-1}) which is assumed to be equal to the solution heat capacity, and MW_X and m_X are the molecular weight and mass of the substance of interest, respectively. In the usual procedure, KCl dissolution in water is used as the recommended standard procedure for which the value of $\Delta_{sol}H_m^o$ is well known ($17.584 \pm 0.017 \text{ kJ mol}^{-1}$ [18]) to calculate C_{CAL} .



In a second experiment, $\Delta_{sol}H_m^o$ is calculated for the substance of interest employing the same calorimeter and the calculated value of C_{CAL} ; it is important that the same experimental conditions are maintained for both experiments.

Such experiments were proposed for the experimental course on Physical Chemistry for Biochemistry majors at the Faculty of Science of the University of the Republic (Uruguay). Students determined the calorimeter's C_p by dissolving an accurately weighed amount of potassium chloride (KCl) in water, whose ΔH is well-known. They subsequently used this C_p value to determine the enthalpies of dissolution of two substances of biochemical interest: urea and guanidinium chloride, two common protein denaturants.

The main experimental issues when carrying out calorimetric analysis are related to the adiabaticity of the physical system and the sensor response speed. The lack of adiabaticity can be inferred from deviations from ideal behavior. Under ideal conditions, the initial and final temperature regions would remain constant (zero slope). In contrast, the regions marked as “a” in Fig. 5 show a positive slope, indicating a net heat flow into the calorimetric system due to heat exchange with the environment. On the other hand, the time delay in reaching the final temperature (marked as “b” in Fig. 5) can be attributed to insufficient agitation, slow dissolution of the substance, or a poor response of the temperature sensor. Additionally, another feature in the thermograms (denoted as “c” in Fig. 5) is attributed to a local overheating or overcooling of the solution in the vicinity of the sensor.

In our design, the agitation is performed by the integrated magnetic stirrer of the commercial thermal container, which operates at a fixed speed defined by the manufacturer. The rotational speed is not specified by the manufacturer and is not user-adjustable. To

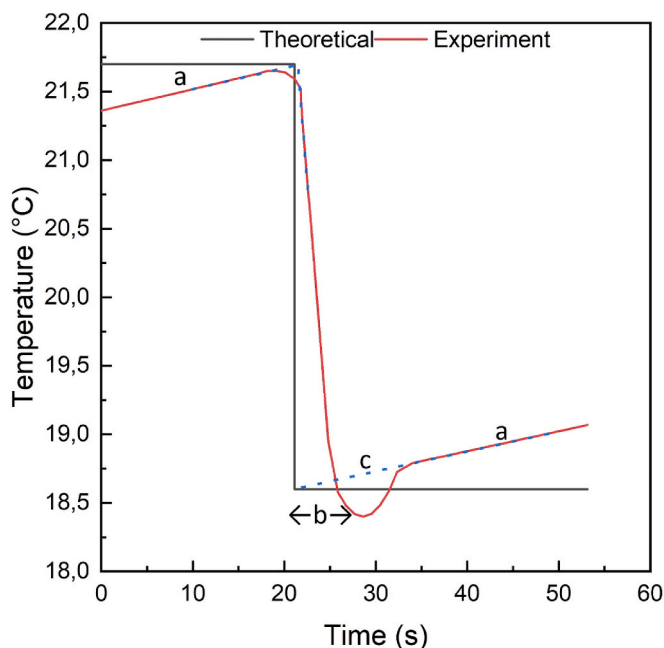


Fig. 5. Theoretical (black line) and experimental (red line) curves for an endothermic process measured in an isobaric calorimeter, manifested as a temperature drop in the surrounding. (a) Region with positive slope due to heat exchange with the environment (lack of adiabaticity). (b) Time delay associated with dissolution kinetics, stirring efficiency, and temperature sensor response. (c) Local overcooling in the vicinity of the temperature sensor. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

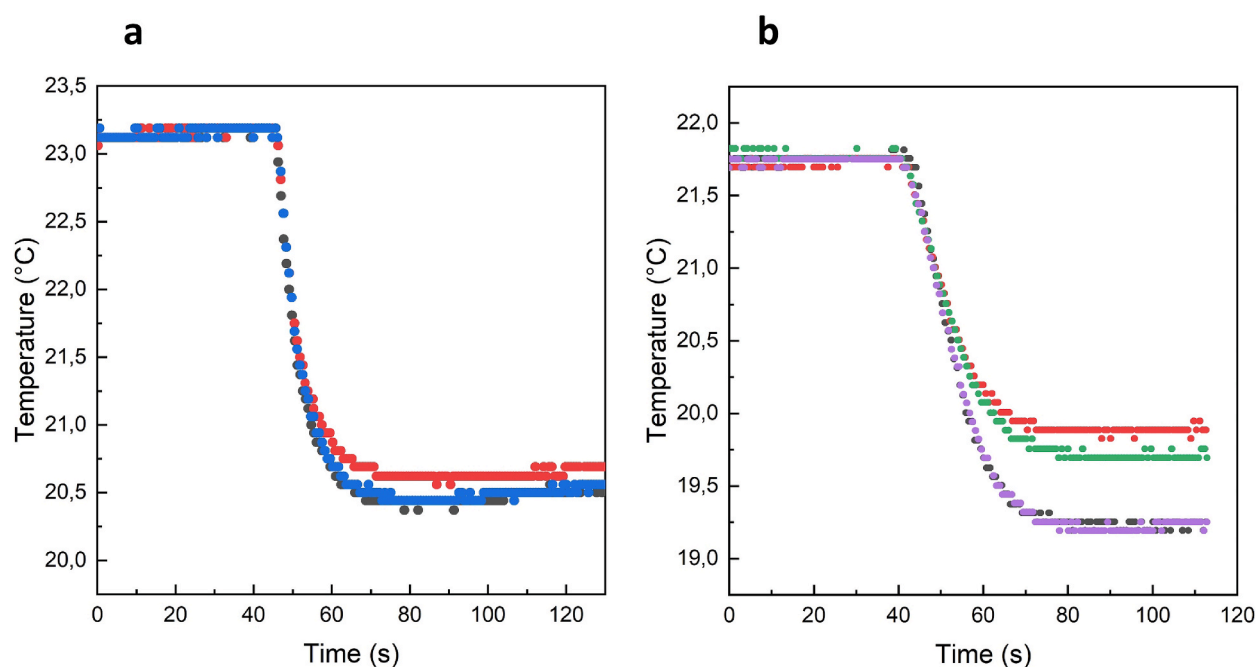


Fig. 6. Experimental data obtained by students for the dissolution enthalpy measurements of (a) guanidinium chloride (GdnHCl: 9.66 g, red; 9.70 g, black; 9.72 g, blue) and (b) urea (5.87 g, red; 5.92 g, green; 7.19 g, purple; 7.49 g, black). Different colors correspond to independent experimental runs performed under otherwise identical conditions. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

provide an order-of-magnitude estimate of the stirring speed, the rotation of the magnetic impeller was evaluated by high-speed video analysis (240 fps) using a visual reference mark. Frame-by-frame analysis yielded a stirring rate on the order of $4\text{--}5 \times 10^3$ rpm under typical operating conditions. Nevertheless, this level of agitation is sufficient to ensure rapid thermal homogenization of the solution and effective dissolution of the substance. As derived from Fig. 6, no evidence of lack of adiabaticity or overcooling can be seen, and the magnetic agitation is adequate to ensure thermal equilibrium in the time lapse of the experiments. However, a time delay of ca. 30 s is evident and can be attributed to a combination of factors, such as the sensor response speed and the time required for the chemical reaction and the system to reach thermal equilibrium.

For the experimental determination of C_{CAL} , the maximum value of the associated uncertainty ($\sigma_{C_{CAL}}$) can be calculated as the square root of the quadratic sum of all variables Y_i , as shown in Eq. (3).

Table 2

Representative dataset used for calorimeter calibration and determination of dissolution enthalpies using Eq. (2).

Experimental values	Values (units)	Result (from Eq. (2))
Calibration with KCl		
m_{KCl}	7.34 ± 0.02 g	$C_{CAL} = 76 \pm 4$ J °C ⁻¹
MW_{KCl}	74.55 g mol ⁻¹	
m_w	147.45 ± 0.02 g	
ΔT	-2.5 ± 0.1 °C	
Enthalpy of dissolution of urea		
m_{urea}	7.19 ± 0.02 g	$\Delta_{sol}H_m^0(urea) = 14.5 \pm 0.9$ kJ mol ⁻¹ (reference: 15.38 ± 0.03 kJ mol ⁻¹) [20]
MW_{urea}	60.06 g mol ⁻¹	
m_w	147.87 ± 0.02 g	
ΔT	-2.5 ± 0.1 °C	
C_{CAL}	76 ± 4 J °C ⁻¹	
Enthalpy of dissolution of guanidinium chloride (GdnHCl)		
m_{GdnCl}	9.70 ± 0.02 g	$\Delta_{sol}H_m^0(GdnHCl) = 18 \pm 1$ kJ mol ⁻¹ (reference: 18.87 ± 0.04 kJ mol ⁻¹) [19]
MW_{GdnCl}	95.53 g mol ⁻¹	
m_w	141.48 ± 0.02 g	
ΔT	-2.8 ± 0.1 °C	
C_{CAL}	68 ± 3 J °C ⁻¹	

$$\sigma_{C_{CAL}} = \sqrt{\sum_i \left(\frac{\partial C_{CAL}}{\partial Y_i} \sigma_{Y_i} \right)^2} \quad (3)$$

where σ_{Y_i} is the uncertainty of each variable Y_i . In our experiments, the uncertainties for MW_X and c_w can be disregarded, and using typical experimental values, it can be deduced (not shown) that the main source of experimental error is the determination of the ΔT . However, the measurements carried out by our students on different days using the same calorimeter yielded a coefficient of variation of approximately 10%. The mean enthalpy values calculated by the students were $\Delta_{sol}H_{GndHCl}^0 = (17.7 \pm 0.3) \text{ kJ mol}^{-1}$ ($n = 3$) and $\Delta_{sol}H_{Urea}^0 = (14 \pm 1) \text{ kJ mol}^{-1}$ ($n = 4$) for guanidinium chloride and urea, respectively. Within experimental uncertainty, the calculated dissolution enthalpies are in agreement with reported values of $18.87 \text{ kJ mol}^{-1}$ [19] and $15.38 \text{ kJ mol}^{-1}$ [20] for guanidinium chloride and urea, respectively. Representative experimental data and the corresponding dissolution enthalpies calculated using Eq. (2) are summarized in Table 2.

Considering the system's total heat capacity ($\approx 700 \text{ J } ^\circ\text{C}^{-1}$ using 150 g of water), the observed baseline stability, and the digital sensor resolution ($0.0625 \text{ } ^\circ\text{C}$), the estimated detection limit of the calorimeter is on the order of 100 J. This corresponds to a temperature change of $\approx 0.14 \text{ } ^\circ\text{C}$, which is approximately twice the sensor resolution and represents the smallest temperature variation that can be reliably distinguished in practice.

8.1. Concluding remarks

We presented a low-cost, open-hardware isobaric calorimeter based on a repurposed commercial thermal mug, an Arduino UNO board, and open source software. The device enables automated temperature acquisition, real-time visualization, and data export, providing a robust and reproducible platform for teaching thermochemistry in undergraduate laboratory courses.

Validation experiments demonstrate that the system yields dissolution and neutralization enthalpies consistent with literature values, while maintaining simplicity, low cost, and ease of replication. Although the developed calorimeter does not aim to replace research-grade commercial instruments, it effectively bridges the gap between rudimentary educational kits and expensive professional equipment. The open architecture of the hardware and software further allows customization and extension, supporting interdisciplinary training in chemistry, electronics, and laboratory automation.

8.2. Safety considerations

The device operates at low voltage (5 V) and does not involve hazardous materials or open electrical components. It poses no specific safety risks beyond standard laboratory handling.

CRediT authorship contribution statement

Marcel Bentancor: Writing – original draft, Software, Resources, Methodology. **Mauricio Ávila:** Validation, Investigation. **Sabrina Maccio:** Validation, Investigation. **Eduardo Méndez:** Writing – review & editing, Resources, Investigation, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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